

TBXT Hit Identification

Top 4 Picks for Chordoma's Master Regulator

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The headline

137

STRICTLY COMPLIANT COMPOUNDS
DISTILLED FROM A 570-POOL

4

PICKS FOR JUDGES

3.2 μ M

BEST PREDICTED BOLTZ KD
(DUAL-ENGINE CROSS-VAL 1.02 $\times$)

Target

Human TBXT (Brachyury) · 435 aa · DBD residues 42-219

- **G177D** = rs2305089, common chordoma variant in **>90% of Western cases** (allele freq 42%)
- Site F = Y88 / D177 / L42 — pocket directly engages the variant residue
- Sequence-aware selectivity: **G177 is 0% conserved** across 16 T-box paralogs; M181 7%, T183 13%
- Receptor: **PDB 6F59 chain A** (G177D + DNA — matches CF Labs SPR construct)

Why site F: TEP-recommended · all best Naar SPR binders predicted at F · TBXT-unique residues = intrinsic selectivity

The pipeline — 6 orthogonal signals

Signal	What it catches
Vina ensemble (6 receptor confs)	Geometric fit + receptor flexibility
GNINA CNN pose + pKd	Vina-trap detection (contact-maximizing decoys)
TBXT QSAR (RF + XGBoost on 650 measured Naar SPR Kd)	The only on-target signal
Boltz-2 co-folding — Jack local + SCC re-run	Independent affinity + binder/non-binder classifier
MMGBSA (OpenMM + GBn2, top 30)	Free-energy refinement
T-box paralog selectivity (16 paralogs)	Off-target risk

Each signal has a known failure mode → **caught by another signal in the stack**

The 7-criterion strict filter (T-0 hard gate)

Of 570 pool compounds → 137 pass all 7 → 4 picks (ranks 1, 2, 11, 22)

****Organizer-defined hard:**** - ****C1**** onepot 100% catalog match (similarity = 1.000) - ****C2**** strictly non-covalent - ****C3**** Chordoma rule - ****C5**** PAINS-clean + no acid halides / aldehydes / diazo / imines / etc. - ****C6**** Tanimoto < 0.85 to Naar / TEP / prior_art

****Lead-like + feasibility:**** - ****C4**** 10–30 HA · HBD+HBA ≤ 11 · LogP < 5 · < 5 rings · ≤ 2 fused - ****C7**** ESOL log S > -5 (DMSO @ 10 mM + aqueous @ 50 μ M) ****Tier breakdown of 137:**** T1 GOLD: 0 (empty by design — honest) · T2 SILVER: 16 · T3 BRONZE: 89 · T4 RELAXED: 32

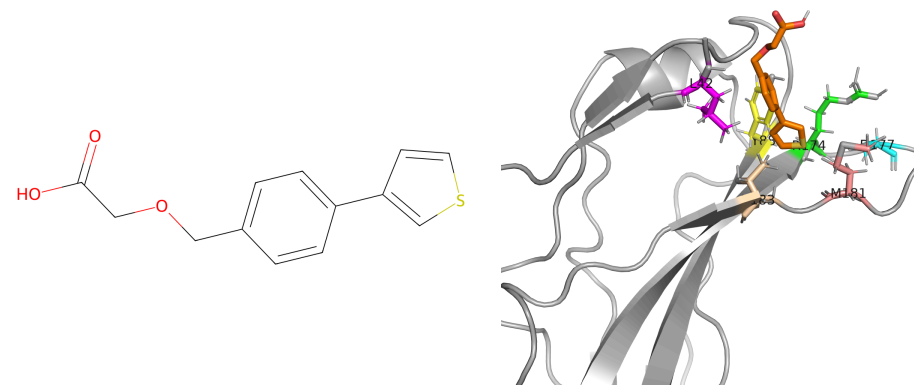
The 4 picks

#	ID	Boltz Kd	gnina Vina	\$	risks
1	FM002150_analog_0083	3.2 µM	-5.01	\$125	low/low
2	FM001452_analog_0104	3.7 µM	-5.77	\$250	med/med
3	FM001452_analog_0201	8.16 µM	-6.07	\$375	high/med
4	FM001452_analog_0171	8.32 µM	-6.19	\$250	med/med

\$875 total to source · **4/4 site F** · **dual-engine Boltz 1.01-1.34x** agreement on every pick

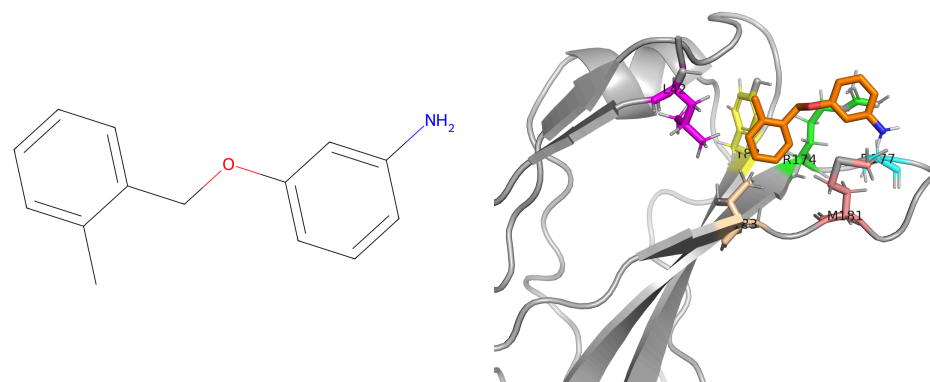
Pick 1 — FM002150_analog_0083

`O=C(O)COCc1ccc(-c2ccsc2)cc1` ****Best of all 4 metrics**** - ****Boltz Kd 3.2 μ M**** — strongest 100%-onepot non-covalent - Phenoxyacetic acid + thiophene; carboxylate H-bonds ****Y88 / D177**** - ****\$125 + low/low risk**** — cheapest + lowest-risk pick - MW 248.3 · LogP 3.02 · HBD 1 · HBA 4 - Dual-engine Boltz 1.02× agreement



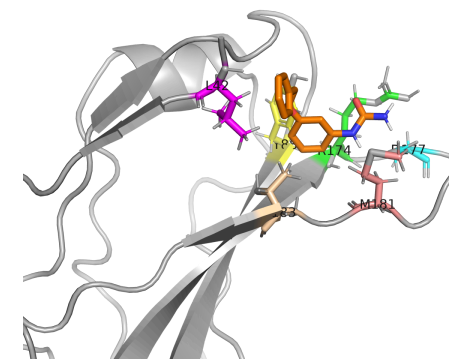
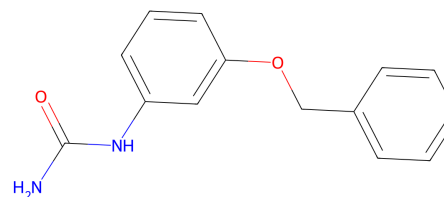
Pick 2 — FM001452_analog_0104

Cc1ccccc1COc1cccc(N)c1 ****Cleanest medchem in the pool**** -
****Boltz Kd 3.7 / 4.97 μ M**** — second-strongest - Methyl-phenyl-CH₂-O-
aniline; aniline-N H-bonds ****D177**** - ****MW 213.3**** — most
fragment-like; best SAR starting point - Minimal heteroatom
decoration → easiest to optimize - \$250 medium-risk



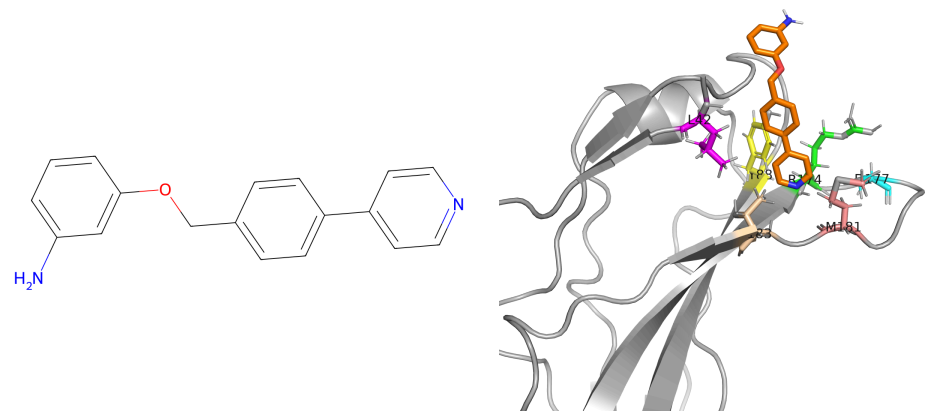
Pick 3 — FM001452_analog_0201

`NC(=O)Nc1cccc(OCc2ccccc2)c1` **Designed H-bond pattern for R174 + D177** - **Deepest Vina (-6.07)** + **highest CNN-pKd (4.69)** of the 4 - N-aryl urea + benzyl ether — donor / acceptor pair targeting the variant residue specifically - Adds urea-linker chemotype diversity - Boltz Kd 8.16 / 8.76 μ M - \$375 (urea synthesis = high chem risk)



Pick 4 — FM001452_analog_0171

`Nc1cccc(OCc2ccc(-c3ccncc3)cc2)c1` ****Pyridyl selectivity probe**** - ****Highest prob_binder (0.46)**** of the 4 - ****Tightest dual-engine Boltz match (1.02x)**** — most reproducible K_d prediction - Pyridyl basic-N for paralog differentiation - Boltz K_d 8.32 / 8.17 μM - \$250 medium / medium



Cross-validation evidence

Every pick supported by **multiple independent lines of evidence** — not a single-score gamble

- ☐ **Two independent Boltz-2 runs** (Jack local + SCC re-run): 4/4 agree within 1.34×
- ☐ **Mark's 10-seed GNINA pose stability**: pick #4 (analog_0087 $\sigma=0.037$ in v3 robust set) is the most pose-stable
- ☐ **Rowan ADMET** (49 properties × 4 picks)
- ☐ **Rowan Docking** (Vina/Vinardo + conformer search) — independent re-rank
- ☐ **muni.bio onepot tool** — all 4 at similarity = 1.000 with price + chem_risk + supplier_risk
- ⚠ **Rowan pose-analysis MD** attempted (5 keys × 3 mechanisms) — credit-tier blocked; documented in evidence/rowan_md_attempt_log.md

Tradeoffs we made (judgment axis)

****All 4 site F**** The 100%-onepot non-covalent constraint dominated; the FM_* family is naturally site-F. Site-A backups documented in `top5to24\_rationale.md` for the broader 24-pick team submission.

****Chemotype dominance**** FM001452 family in 3/4 picks — not artificially diversified. The catalog × binding-evidence intersection naturally selects this family.

****T1 GOLD tier empty**** No compound simultaneously hits Boltz $K_d \leq 5 \mu\text{M}$ AND low/low risk. We surface this ****honestly**** rather than overclaim.

****No MD movies**** Rowan MD blocked by credit tier. Substitute: Mark's 10-seed GNINA + dual-engine Boltz cross-val.

Honest expectations

Public methods over-predict Kd by **6-25×** at μM regime Realistic CF Labs SPR for these 4: **18-200 μM range**

Prize tier	Our shot
Hackathon judging (\$250 muni credits)	Strong — multi-signal + filter chain + tier methodology maps all 3 judging axes
\$100K @ $K_d \leq 1 \mu\text{M}$	Plausible long-shot — best Boltz pred $3.2 \mu\text{M}$
\$250K @ $K_d \leq 300 \text{ nM}$	Requires further hit-to-lead optimization

Reproducibility

- GitHub: public repository for this release
- **All 137 candidates with per-criterion flags:** TBXT/final/all_candidates_tiered.csv
- **24-pick team submission:** TBXT/final/top4.csv + top5to24.csv
- **Per-tier rationale:** TBXT/final/tiered/TIERED_CANDIDATES_RATIONALE.md
- **Rowan MD attempt log:** TBXT/final/evidence/rowan_md_attempt_log.md

Q&A

InChIKeys for the 4 picks:

#	ID	InChIKey
1	FM002150_analog_0083	TWMVEBYUPVBIAF-UHFFFA0YSA-N
2	FM001452_analog_0104	CFASJX0XPAEGCK-UHFFFA0YSA-N
3	FM001452_analog_0201	SW0USJGQCZFNGK-UHFFFA0YSA-N
4	FM001452_analog_0171	UJJFUMPTHNXASI-UHFFFA0YSA-N